

5.2 Polar Worksheet

1. A polar point has the coordinates. Find and fully simplify the Cartesian coordinates of this point.
2. A Cartesian point has the coordinates $(5, \sqrt{75})$. Find a polar coordinate representation for this point that has a positive value for r and another representation that has a negative value for r .
3. Convert $\left(6, \frac{\pi}{3}\right)$ to rectangular coordinates.

4. Convert $(2, -2)$ to polar coordinates.

5. Convert $(4, 3\pi)$ to rectangular coordinates.

6. Convert $(0, 3)$ to at least 3 different points in polar coordinates.

7. Convert $\left(-4, -\frac{\pi}{4}\right)$ to rectangular coordinates.

8. Convert $(-\sqrt{3}, -1)$ to polar coordinates.

9. Sketch a graph of the continuous polar curve whose points satisfy the following criteria.

- As θ increases from 0 to $\pi/2$, r decreases from 4 to 2.
- As θ increases from $\pi/2$ to π , r decreases from 2 to 0.
- As θ increases from π to $3\pi/2$, r decreases from 0 to -1.
- As θ increases from $3\pi/2$ to 2π , r decreases from -1 to -3.

10. Sketch a graph of the continuous polar curve whose points satisfy the following criteria.

- As θ increases from 0 to $\pi/2$, r increases from 1 to 3.
- As θ increases from $\pi/2$ to π , r decreases from 3 to 1.
- As θ increases from π to $3\pi/2$, r increases from 1 to 3.
- As θ increases from $3\pi/2$ to 2π , r decreases from 3 to 1.

11. Sketch a graph of the continuous polar curve whose points satisfy the following criteria.

- As θ increases from 0 to $\pi/2$, r stays constant at -2.
- As θ increases from $\pi/2$ to π , r increases from -2 to 0.
- As θ increases from π to $3\pi/2$, r increases from 0 to 2.
- As θ increases from $3\pi/2$ to 2π , r stays constant at 2.

12. Sketch a graph of the continuous polar curve whose points satisfy the following criteria.

- As θ increases from 0 to $\pi/2$, r decreases from 2 to 0.
- As θ increases from $\pi/2$ to π , r decreases from 0 to -2.
- As θ increases from π to $3\pi/2$, r increases from -2 to 0.
- As θ increases from $3\pi/2$ to 2π , r increases from 0 to 2.

13. Sketch a graph of the continuous polar curve whose points satisfy the following criteria.

- At $\theta = 0$, the curve is at $(-4, 0)$.
- Between $0 < \theta < \pi$, r is always negative and increasing.
- Between $\pi < \theta < 3\pi/2$, r is always positive and increasing.
- Between $3\pi/2 < \theta < 2\pi$, r is always positive and decreasing.
- $r = 0$ only when $\theta = \pi$.
- At $\theta = 2\pi$, the curve is at $(2, 0)$.

14. Convert the polar equation $r = \frac{1}{5 \cos(\theta) + 3 \sin(\theta)}$ to a Cartesian equation and solve this resulting equation for y .

15. Convert the following polar equation to Cartesian coordinates $r = 2 \sec(\theta)$.

16. Convert the following rectangular equation to polar coordinates $x = \frac{3}{y}$.

17. Convert the following polar equation to Cartesian coordinates $r \sin(\theta) = r \sqrt{\cos(2\theta)}$.

18. Convert the following rectangular equation to polar coordinates $y^2 = 5x$.